

Long-Term Changes in Adiposity and Glycemic Control Are Associated With Past Adenovirus Infection

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OBJECTIVE Ad36, a human adenovirus, increases adiposity but improves glycemic control in animal models. Similarly, natural Ad36 infection is cross-sectionally associated with greater adiposity and better glycemic control in humans. This study compared longitudinal observations in indices of adiposity (BMI and body fat percentage) and glycemic control (fasting glucose and insulin) in Ad36-infected versus uninfected adults. **RESEARCH DESIGN AND METHODS** Baseline sera from Hispanic men and women (n = 1,400) were screened post hoc for the presence of Ad36-specific antibodies. Indices of adiposity and glycemic control at baseline and at ~10 years past the baseline were compared between seropositive and seronegative subjects, with adjustment for age and sex. In addition to age and sex, indices of glycemic control were adjusted for baseline BMI and were analyzed only for nondiabetic subjects. **RESULTS** Seropositive subjects (14.5%) had greater adiposity at baseline, compared with seronegative subjects. Longitudinally, seropositive subjects showed greater adiposity indices but lower fasting insulin levels. Subgroup analyses revealed that Ad36-seropositivity was associated with better baseline glycemic control and lower fasting insulin levels over time in the normal-weight group (BMI ≤ 25 kg/m²) and longitudinally, with greater adiposity in the overweight (BMI 25–30 kg/m²) and obese (BMI >30 kg/m²) men. Statistically, the differences between seropositive and seronegative individuals were modest in light of the multiple tests performed. **CONCLUSIONS** This study strengthens the plausibility that in humans, Ad36 increases adiposity and attenuates deterioration of glycemic control. Panoptically, the study raises the possibility that certain infections may modulate obesity or diabetes risk. A comprehensive understanding of these under-recognized factors is needed to effectively combat such metabolic disorders.